

Effective data-driven precision medicine by cluster-applied deep reinforcement learning

Sang Ho Oh^a, Su Jin Lee^b, Jongyoul Park^{a,c,*}

^a Research Center of Electrical and Information Technology, Seoul National University of Science and Technology, Seoul, 01811, Republic of Korea

^b Department of Internal medicine, Seoul Red Cross Hospital, Seoul, 03181, Republic of Korea

^c Department of Applied Artificial Intelligence, Seoul National University of Science and Technology, Seoul, 01811, Republic of Korea

ARTICLE INFO

Article history:

Received 25 April 2022

Received in revised form 24 August 2022

Accepted 6 September 2022

Available online 12 September 2022

Keywords:

Deep reinforcement learning

Precision medicine

Clustering

Recommendation system

Healthcare management

ABSTRACT

The significance of machine-learning approaches in the healthcare domain has grown rapidly owing to the existence of enormous amounts of data and well-established simulation models and algorithms. The digitization of health-related data, as well as rapid technological advancements are accelerating the development and application of machine learning in healthcare, particularly in precision medicine. The ultimate goal of precision medicine is to provide personalized medicine, which requires tailoring medical decisions to each patient based on their projected disease response. In this study, we propose a cluster-applied deep reinforcement learning-based type 2 diabetes treatment recommendation model based on the electronic health records of South Koreans. The purpose of applying a clustering algorithm is to group patients who are in a similar state, to boost the performance of deep reinforcement learning, build a more realistic treatment recommendation model to support clinicians, and develop expert systems in the field of healthcare. The proposed model demonstrated significant performance by decreasing diabetes-related medical checkup measurements. Furthermore, the proposed model delivered high-quality performance when compared with existing reinforcement-learning methods. Finally, the recommendation outcomes of the proposed model were validated against real-life prescriptions to ensure the accuracy of the findings.

© 2022 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Precision medicine is defined as gaining a better understanding of a disease to support diagnosis and prevention, creating tailored medicines, and more precisely forecasting disease outcomes to provide personalized healthcare to a subset of patients. Precision medicine is a trend shifting away from treatment based on guidelines and toward treatment based on the requirements of the individual [1]. The incremental expansion of biomedical data obtained from patients including genetics, medicine, imaging, and metabolism has had a strong influence on the development of precision medicine [2]. One-size-fits-all treatment regimens are likely to be inappropriate in terms of both patient outcomes and resource utilization owing to the varied characteristics of patient disease trajectories [3]. The ultimate goal of precision medicine is personalized medicine, which means tailoring treatment decisions, practices, procedures, and solutions to patients based on their projected disease outcome or uncertainty [1]. Clinical medicine, fundamental biomedical research, and new sources of

health-related data are currently generating massive amounts of biomedical data that require the use of computers to assist in analysis. However, the area of computer science has also advanced, providing additional management and processing options of these massive datasets. Precision medicine diagnostics and therapies can be developed using computers to process the large amounts of data collected in the healthcare domain.

Owing to the existence of enormous amounts of data and well-established simulation models and algorithms, the significance of machine-learning approaches in the healthcare domain has grown rapidly [4–6]. The digitization of health-related data, as well as rapid technological advancements are accelerating the development and application of machine learning in healthcare, particularly in precision medicine [7–11]. Reinforcement learning (RL) is a type of machine learning that has demonstrated significant practical and methodological progress in recent years in terms of representation, generalization, and efficiency [12]. The agent reacts based on its present state, whereas the environment provides input and predicts the next state. The agent's purpose is to select an optimal policy that maximizes the overall reward over time. Consequently, in RL, agents do not receive explicit guidelines on action; instead, they must determine what behavior is superior by exploring the environment.

* Corresponding author at: Department of Applied Artificial Intelligence, Seoul National University of Science and Technology, Seoul, 01811, Republic of Korea.
E-mail address: jongyoul@seoultech.ac.kr (J. Park).

Medical treatment typically comprises a series of actions that determine the potential treatment such as medication scheduling or type and dosage, depending on the present health status of the patient and their clinical history to obtain the maximum long-term benefits. The use of RL approaches to address healthcare application decision-making problems appears to have numerous advantages. RL can make decisions regarding the best treatment for the patient considering patient diversity in treatment response due to variances in disease intensity, personal characteristics, and pharmacodynamics. Moreover, under specific circumstances, RL can acquire optimal policies based on previous treatment and outcome experience. This makes RL more appealing than other simulation-based studies because building a framework that sufficiently depicts the difficulty and variety of biomedical systems and their unique outcomes to treatments is frequently challenging. Furthermore, RL-based methods allow for improved long-term outcomes by considering the delayed result of medicines, which is a key feature of medical therapy [13].

Diabetes mellitus (DM) is a multifactorial chronic illness caused by a combination of hereditary and environmental factors [14,15]. The disease is associated with a high level of familiarity and the prevalence of diabetes differs based on ethnicity. The number of people with diabetes has virtually doubled since 1980, to 422 million; it is expected to reach 693 million by 2045 according to the World Health Organization's global report on diabetes [16]. High levels of glucose in the blood are a symptom of the condition, which is caused by a lack of insulin concentration and/or activity, a pancreatic hormone involved in glycemic control. There is currently no cure for diabetes, although it can be treated and managed with medication and insulin to maintain blood sugar levels as close to normal as feasible and avoid the onset of diabetes complications [17]. Despite the availability of numerous drugs, diabetes has been poorly controlled owing to major knowledge gaps, anti-diabetes therapy adoption, and blood-glucose-level adequacy [18–20]. To solve this problem, experts and machine-learning-based intelligent systems must collaborate for effective health management.

Recently, research has been conducted on diabetes and other disease-treatment recommendation systems using RL. Unlike traditional machine learning methods, RL generates a policy from experience to forecast states and rewards while learning the procedure, instead of requiring a well-represented theoretical modeling approach. Therefore, RL is suited to applications with inherent time delays such as those where decisions are taken without current outcome knowledge, yet are judged based on a longer-term future reward. These characteristics make RL an interesting approach for designing suitable policies in a variety of healthcare applications, where decision-making is typically characterized by a lengthy sequential procedure with delayed feedback [21].

Ref. [22] utilized Q-learning to suggest insulin dosages to control blood glucose levels in diabetic patients considering the patients' state as characterized by glycated hemoglobin levels, physical activity, body mass index (BMI), and alcohol use. Ref. [23] applied deep RL to automated anesthetic dosing using a continuous-action model based on the actor-critic RL paradigm. Ref. [24] utilized a retrospective dataset of patients with type 2 diabetes mellitus (T2DM) to model glycemia, blood pressure, and cardiovascular disease risk as health outcomes. They developed a deep Q-network (DQN) RL method that optimizes treatment regimens. Ref. [25] used a deep deterministic policy gradient (DDPG) algorithm to improve the sepsis AI clinical decision system to help clinicians address changes in the condition of sepsis patients, reduce the workload, and provide a reference dosage. Finally, Ref. [26] proposed treatment recommendation models for T2DM patients based on deep RL, namely double DQN (DDQN), using the electronic health records (EHRs) from the Singapore

Health Service. These studies used algorithms where experts and machine learning-based intelligent systems collaborated to manage good health.

In our research, we aimed to develop a cluster-applied deep RL-based personalized treatment recommendation model for T2DM utilizing a large number of EHRs. Owing to the different therapeutic goals and individual heterogeneity of patients, diabetes therapy can be challenging considering the requirement for precise, personalized treatment. In the early stages of T2DM, metformin (MET) monotherapy has been used to treat the disease [27,28]. When a patient's condition does not improve with monotherapy, the treatment option can be switched to dual or triple therapy [29–31]. Multiple (dual/triple) treatments have been found to be efficient in maintaining adequate glucose levels and preventing complications in previous research [32–35]. The proposed model recommends the use of mono, dual, or triple therapy among T2DM medications, for the patients in each cluster.

The main contributions of our study follow. First, we expressed the patient's state by considering the biomedical data of the patient including demographic information, fasting plasma glucose (FPG), blood pressure, BMI, and other laboratory tests. Secondly, we applied the K-means clustering algorithm to divide patients into similar state groups to boost the performance of the RL algorithm. Lastly, we verified that the proposed model's recommendation results were reasonable and appropriate when compared with real-life experts' prescriptions and evaluated the effectiveness of the proposed T2DM treatment recommendation model by decreasing FPG levels and the occurrence rate of diabetes-related complications among those who complied with the recommendation of the model. Furthermore, we validated the performance of the proposed model by comparing it with existing RL algorithms from previous studies, such as Q-learning, DQN, DDQN, DDPG, and actor-critic.

2. Methodology

2.1. Data descriptions

The EHRs used in this study were obtained from the National Health Insurance Sharing Service (NHIS) of Korea. The NHIS is a national institute that manages the Korean national health insurance program. They collect health data for research purposes according to the applicable laws and regulations. This includes obtaining the informed consent from all participants (if they are under the age of 18, consent is obtained from a parent or legal guardian). The database covers the period 2003–2013.

The NHIS manages records of healthcare utilization, laboratory test results, and prescriptions. Laboratory test results include results from medical check-ups and habitual and behavioral data from the questionnaire. Specifically, age, weight, height, FPG, total cholesterol (TC), blood pressure, high-density lipoprotein and low-density lipoprotein cholesterol, triglycerides, smoking status, drinking and exercise frequency, and history of diabetes, heart disease, stroke, and hypertension are recorded.

Because we required patients with T2DM, we selected patients based on the following criteria. First, we filtered out patients diagnosed with T2DM based on the International Classification of Diseases (ICD)-10 codes, which are E10 to E14. Then, we selected patients who had received T2DM prescriptions for more than half of their disease period. Lastly, we sorted patients who had complete laboratory test results from their appearance to the end of the data period.

The total number of T2DM patients after the filtering process was 22,581. All participants were tracked from the beginning of 2003 to the date of death, or December 31, 2013, whichever occurred first.

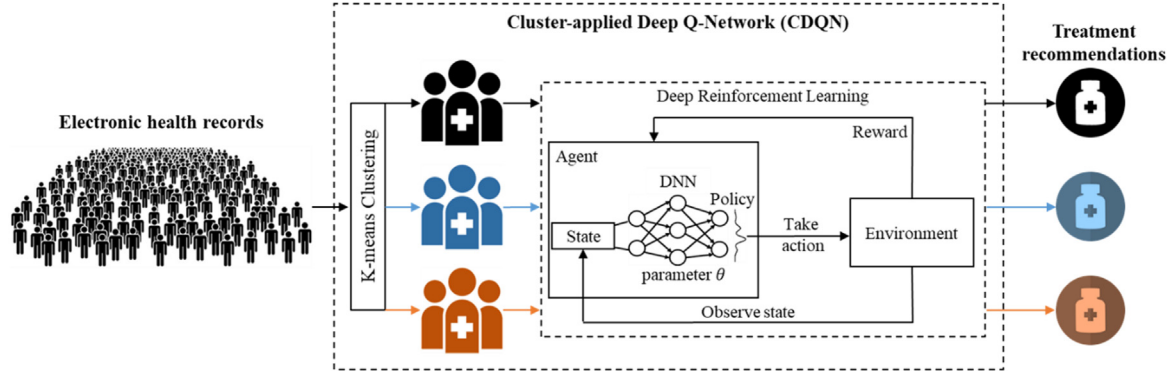


Fig. 1. Research framework.

Table 1
Description of dataset.

Category	Male	Female
Sex (%)	54	46
Age, mean (SD)	58 (34)	65 (28)
T2DM period (years), mean (SD)	5.4 (3.8)	6.2 (2.6)
FPG (mg/dL), mean (SD)	142.5 (41.8)	147.1 (48.3)
BMI (kg/m ²), mean (SD)	25.7 (7.2)	26.7 (6.1)
Systolic blood pressure (mmHg), mean (SD)	134.2 (38.7)	139.7 (34.8)
Diastolic blood pressure (mmHg), mean (SD)	86.7 (21.5)	88.3 (23.4)
TC (mg/dL), mean (SD)	194.2 (48.3)	197.2 (52.8)
Smoker (%)	62	43
Family history of T2DM (%)	32	35

SD: Standard deviation.

Table 1 describes the 22,581 T2DM patients who were analyzed. With mean ages of 58 and 65 years, male and female patients accounted for 54 and 46 percent of the data, respectively. Male and female T2DM patients had T2DM for an average of 5.4 and 6.2 years, respectively. The average BMI of both sexes was overweight and average FPG levels were greater than 140 mg/dL, which is critical for T2DM control. The amount of TC in the blood was virtually normal. Both male and female patients had high blood pressure. Male and female patients smoke at rates of 62 percent and 43 percent, respectively. Lastly, T2DM ran in the family for 32 percent of the men and 35 percent of the women.

2.2. Cluster-applied DQN (CDQN)

To improve the performance of deep RL, this study used the K-means algorithm to cluster T2DM patients into similar state groups. Then, based on their current laboratory test result metrics, we used a DQN to recommend for each clusters' medication treatment. Fig. 1 illustrates the research framework.

2.2.1. K-means clustering

K-means clustering is an iterative clustering algorithm that is both simple and effective [36]. K-means clustering partitions n objects into k clusters, where each object belongs to the cluster with the nearest mean. This method produces exactly k different clusters with the greatest possible distinction. The objective of K-means clustering is to minimize total intra-cluster variance, or, the squared error function represented as (1) by Euclidean distance chosen as the index, given dataset X having n dimensional points and k classes to be partitioned [37].

$$d = \sum_{k=1}^k \sum_{i=1}^n \|(x_i - u_k)\|^2, \quad (1)$$

where k is the number of clusters that have a direct influence on the clustering results [38], n is the number of cases, x_i is the i th point in the dataset, and u_k is the centroid of cluster k . In our study, we set $k = 3$, which means that three clusters were generated.

As K-means clustering tends to converge on an optimal set of cluster centers and cluster participants based on the distance from these centroids through successive iterations, it would appear that the more optimally placed these initial centroids are, the fewer iterations of the K-means clustering algorithms necessary for convergence. This implies that the initialization of these initial centroids could be useful. In this study, we utilized the K-means++ method. To spread the initial centroids, K-means++ assigns the first centroid to the location of a randomly chosen data point. The subsequent centroids are then selected from the remaining data points with a probability proportional to the square of the distance from the closest existing centroid of a given point. The outcome is an attempt to spread the centroids as much as possible by utilizing all of the occupied data space as of initialization [39].

2.2.2. DQN

In certain situations, such as the state variable's transition function or the probability distribution is known, Q-learning is sufficient for selecting the appropriate course of action. The foundation of Q-learning is the estimation of a value function, which is a set of Q-values. The Q-learning approach [22] estimates the Q-values for each state-action (s_t, a_t) combination. When the optimal Q-values are converged, the state of the environment must be known to select the optimal action. At the start of the process, the Q-values are set to an arbitrary real number. At iteration t , the agent evaluates the reward value for each (s_t, a_t) combination. The essence of the algorithm is represented in (2), where the Q-values are updated iteratively as a function of the immediate reward r_t and the Q-values of the next state-action combination $Q(s_{t+1}, a_{t+1})$. With a range of 0 to 1, γ is the discount factor that controls the influence of future rewards on the current rewards.

$$Q(s_t, a_t) \leftarrow r_t + \gamma \max_{a_{t+1}} \{Q(s_{t+1}, a_{t+1})\} \quad (2)$$

Regardless of policy, the Q-values are updated such that the sequence of Q-values converges to an ideal action-value function Q^* [40]. One of the most intriguing aspects of Q-learning is its ability to generate state-action pairs using a variety of sampling procedures. The ϵ -greedy action selection, specified in (3), is a typical sampling approach, where $\epsilon \in (0, 1)$.

$$a_t = \begin{cases} \arg \max_{a \in A} Q(s_t, a) & \text{with probability } 1 - \epsilon, \\ a \sim A & \text{otherwise.} \end{cases} \quad (3)$$

Table 2
Diabetes-related complications.

Types of complication	Disease
Chronic complications	Diabetic retinopathy
	Diabetic cataract
	Diabetic nephropathy
	Chronic kidney disease
	Foot disease
	Diabetic neuropathy
Acute complications	Myocardial infarction
	Heart failure
	Hypoglycemia

A function approximator, such as a deep neural network (DNN), can be used as follows: $Q(s, a) = Q(s, a; \theta)$ as an alternative to the Q-table [41,42]. The parameter set of the approximator is θ and the Q-network is $Q(s, a; \theta)$, which is then optimized to approximate (2) [42,43]. There are two networks in a DQN: Q-network $Q(s, a; \theta)$ and a target Q-network $\hat{Q}(s, a; \theta^-)$ [41,44]. The Q-network is used to determine the optimal action and the target Q-network to generate a target value to update θ of $Q(s, a; \theta)$. $Q(s, a; \theta)$ is updated to minimize the mean-square error with $\arg \max_{a'} \hat{Q}(s', a'; \theta^-)$ at every iteration t by optimizing a loss function using (4).

$$L_t(\theta_t) = E \left[(r_t + \gamma \arg \max_{a'} \hat{Q}(s', a'; \theta^-) - Q(s, a; \theta_t))^2 \right] \quad (4)$$

Exploration assesses the potential actions in a DQN, whereas exploitation draws on previous experience. Exploration versus exploitation is a trade-off that must be properly considered. A DQN also employs replay memory, which is an experience replay that saves the agents' experience in a dataset [44]. To update the Q-networks, experience samples are randomly selected from the replay memory [45].

2.2.2.1. State. The state represents the health status of T2DM patients. In our study, we presented five factors that influence the T2DM status, as indicated in (5).

$$S = (S_{AGE}, S_{FPG}, S_{COMPLICATIONS}, S_{PERIOD}, S_{RISK}) \quad (5)$$

S_{AGE} represents age. We denote a patient's age as "1" if he or she is greater than 55 years old; otherwise, we denote it as "0". The age threshold was set at 55 years because individuals over 55 years of age had a greater chance of developing diabetes [46].

S_{FPG} represents the FPG level. We denote it as "0" if the level is less than 100 mg/dL, "1" if it is between 100 mg/dL and 125 mg/dL, and "2" if it is greater than 125 mg/dL. Normal, Warning, and Diabetes stages were assigned to these three levels, respectively.

$S_{COMPLICATIONS}$ represent the diabetes-related complication status. If a patient has one or more of the diabetes-related complications listed in Table 2, we denote the state as "1"; otherwise, it is "0".

S_{PERIOD} represents the T2DM period from diagnosis to the end of the records. If the period is less than four years, we set this value to "0". If the period ranges from four to less than eight years, we set this to "1"; otherwise, it is set to "2".

S_{RISK} represents a patient's risk state as determined by the diabetes risk score calculator [47]. The diabetes risk score calculator was developed as a diagnostic tool to identify people who are at high risk for diabetes and to increase awareness regarding risk factors that could be changed by healthy lifestyles [48]. Demographic information, BMI, family history, smoking status, hypertension, and steroid medication intake are the criteria that are considered when calculating the score. If a patient's score is greater than the threshold score (20 was used as the threshold

Table 3
Summary of state space.

State	Description	Criteria
S_{AGE}	Age of patient	0: Less than 55 1: Equal or greater than 55
S_{FPG}	Fasting plasma glucose level	0: Normal 1: Warning 2: Diabetes
$S_{COMPLICATIONS}$	Diabetes-related complication	0: Without complication 1: With complications
S_{PERIOD}	T2DM period (years)	0: [0,4) 1: [4,8) 2: [8,11)
S_{RISK}	Diabetes risk score of patient	0: Without risk 1: With risk

Table 4
Action space.

Type	Action no.	Description
Monotherapy	1	MET
	2	SUL
	3	Other monotherapies
Dual therapy	4	MET + SUL
	5	MET + DPP4i
	6	Other dual therapies
Triple therapy	7	MET + SUL+ AGI
	8	Other triple therapies

score, which is the median value of the risk score in the data set), the patient received a "1"; otherwise, they received a "0".

The total number of states used in this study was 72 ($2 \times 3 \times 2 \times 3 \times 2$). A summary of these states is presented in Table 3.

2.2.2.2. Action. Action represents the oral diabetes medication prescribed to the patients. In our dataset, there were eight possible diabetes medications: MET, sulfonylureas (SUL), DPP-4 inhibitors (DPP4i), meglitinide (MEG), GLP-1 receptor agonists (GLP-1RA), thiazolidinedione (TZD), alpha-glucosidase inhibitors (AGI), and insulin. In Fig. 2, we arrange the diabetes medications based on the prescription distribution in the dataset. The actions were categorized as mono-, dual-, and triple-therapy.

We selected the first two medications for each therapy and grouped the remaining medications. However, we selected only one action (number 7) for triple therapy because the prescription frequencies of the other triple therapies were insufficient to be considered as an individual action. Therefore, in our research, we selected eight actions based on the prescription frequency, as indicated in Table 4. Monotherapy is typically used for the initial treatment of diabetes, whereas dual and triple therapies are used in patients with severe diabetes.

2.2.2.3. Rewards. Quality-adjusted life-year (QALY) was utilized to evaluate the quality of life while accounting for the disutility caused by medical interventions such as complications or an increase in FPG levels. The QALY is a unit of measure used in economic evaluations to assess the value of medical therapies. The range of QALY is zero to one; a value of one specifies a year of perfect health, whereas zero indicates death. Using this information, the QALY can be the basis for decisions regarding health insurance coverage, treatment options, insurance plan evaluations, and future-plan priorities [49].

The reward function in the proposed model is composed of decrement factors based on the states to apply the concept of

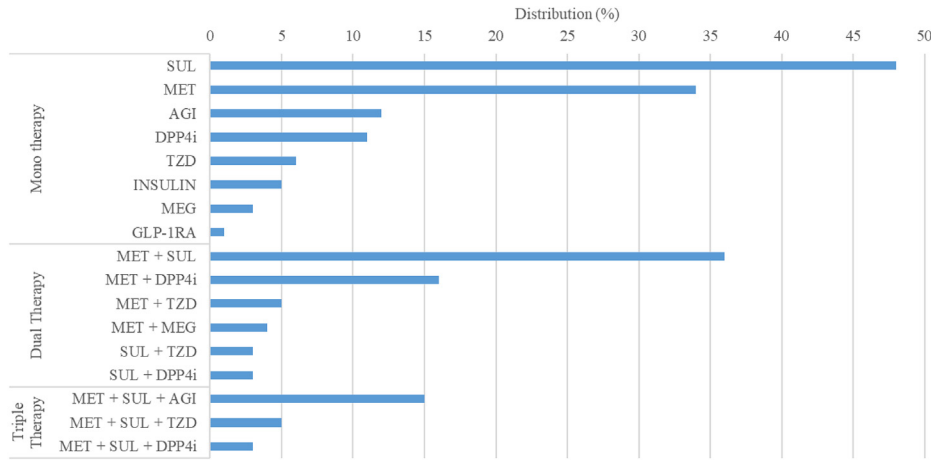


Fig. 2. Distribution of diabetes medication in dataset.

Table 5
Reward values.

Notations	Descriptions	Decrement values
$d^{AGE}(s')$	Utility decrement value of age	<55: 0.06 ≥55: 0.136
$d^{FPG}(s')$	Utility decrement value of FPG level	Normal: 0.035 Warning: 0.09 Diabetes: 0.146
$d^{COMPLICATIONS}(s')$	Utility decrement value of diabetes-related complications	With: 0.235 Without: 0
$d^{PERIOD}(s')$	Utility decrement value of period	[0,4]: 0.081 [4,8]: 0.095 [8,11]: 0.108
$d^{RISK}(s')$	Utility decrement value of risk	With: 0.071 Without: 0
$C^{MED}(a)$	Medication prices	Depends on action

The utility decrement values were referenced from previous studies [51–53].

QALY, as indicated in (6).

$$r_t = r^{WTP} \left[\frac{(1 - d^{AGE}(s'))(1 - d^{FPG}(s'))(1 - d^{COMPLICATIONS}(s'))}{(1 - d^{PERIOD}(s'))(1 - d^{RISK}(s'))} \right] - C^{MED}(a), \quad (6)$$

where r_t is the reward function, a is an action, and s' is its next state. r^{WTP} represents the value that is willing to be paid to maintain one QALY. To calculate the reward, we considered the seven decrement factors to be the same as those of the states. $C^{MED}(a)$ is the medication price [50]. Table 5 displays the descriptions of the reward factors.

3. Discussions of treatment recommendation results and validations

3.1. Results of clustering

In this section, we present the results of the k-means clustering. We set $k = 3$, which causes the algorithm to create three clusters. Table 6 presents the characteristics of the three clusters.

Cluster 1 (C1) included patients with minor T2DM. In C1, patients had normal FPG levels, normal BMI levels, a short T2DM period, and no diabetes-related complications. Cluster 2 (C2) included patients with severe T2DM. They were primarily greater than 55 years old, with high FPG and BMI levels, with a T2DM period longer than in C1 patients. Cluster 3 (C3) patients were

similar to C2 patients; however, C3 patients had more than one diabetes-related complication and high blood pressure. A detailed description of each cluster is presented in Table 7.

Table 7 describes the 22,581 T2DM patients divided into three clusters. The populations of C1, C2, and C3 were 5418, 9512, and 7651 individuals, respectively. The mean (mean of male and female) age of C1 was 48 years, whereas C2 and C3 were 60 and 57 years, respectively. The T2DM period for C1 was relatively short compared to that for C2 and C3. C2 patients had the highest FPG and BMI levels, whereas C3 patients had the highest blood pressure levels. Because only C3 patients had diabetes-related complications, the mean number of complications for C1 and C2 was zero; C3 had 3.2 and 2.4 complications for men and women, respectively.

In Table 7, we present the descriptions of the clusters in males and females to provide a detailed view of the results. However, in our analysis, we analyzed the results by cluster to validate the difference in treatment recommendations between each cluster and to demonstrate the performance of the proposed CDQN.

3.2. Concordance rate validation

Before proceeding to further analysis, it was essential to validate that the CDQN recommendations for T2DM treatment were reasonable and appropriate. To verify that the recommendations were similar to real-life prescriptions, we evaluated the concordance rate between the CDQN recommendations and expert prescriptions. To determine this rate, we evaluated whether the CDQN recommendations and expert prescriptions in each state were identical. The most frequently used medicine in each state was compared to the recommendation of the proposed CDQN's results for expert prescriptions. We computed the concordance rate by counting the number of matched states among the 72 states. Table 8 displays the sample states of the patients for reference.

Table 8 displays sample States 1, 25, and 72. A patient in State 1 was less than 55 years of age, had a Normal FPG level, no diabetes-related complications and risk, and a short T2DM period. For state 25, a patient was greater than or equal to 55 years old, with a Diabetes FPG level, without diabetes complications, without diabetes risk, and short T2DM period. Lastly, patients in state 72 were greater than or equal to 55 years old, with a Diabetes FPG level, with diabetes complications and risk, and a long T2DM period.

We also demonstrated the sample treatment recommendation results for each cluster and how the count matched between the

Table 6
Characteristics of clusters.

Cluster 1. Minor T2DM	Cluster 2. Severe T2DM	Cluster 3. T2DM with Complications
Normal FPG level	Mostly age above 55	High blood pressure
Normal BMI level	High FPG level	High FPG level
Short T2DM period	High BMI level	Average T2DM period
No Complications	Long T2DM period	More than 1 diabetic complication

Table 7
Description of clusters.

Category	C1		C2		C3	
	Male	Female	Male	Female	Male	Female
Population	2438	2980	5327	4185	4429	3222
Age, mean	45	51	57	63	54	60
T2DM period (years), mean	2.1	2.8	5.2	6.5	4.8	5.3
FPG (mg/dL), mean	138.8	139.4	157.5	158.7	152.1	154.5
BMI (kg/m ²), mean	20.4	19.3	29.7	26.2	26.3	27.8
Systolic blood pressure (mmHg), mean	122.5	124.6	135.7	138.4	155.4	157.9
Diastolic blood pressure (mmHg), mean	80.1	81.7	86.3	87.4	91.9	94.2
Number of complications, mean	0	0	0	0	3.2	2.4

Table 8
Sample state of patients.

State no.	Age		FPG			Complications		Period			Risk	
	<55	≥55	Normal	Warning	Diabetes	Without	With	[0,4)	[4,8)	[8,11)	Without	With
1	0		0			0		0			0	
25		0			0	0		0			0	
72		0			0		0			0		0

Table 9
Examples of treatment recommendation results of CDQN.

State no.	CDQN recommendation results			Experts' prescription			Match		
	C1	C2	C3	C1	C2	C3	C1	C2	C3
1	1			1			0		
25		4			4			0	
72			7			7			0

CDQN's results and experts' prescriptions. This is presented in [Table 9](#).

[Table 9](#) indicates the CDQN treatment results, expert prescriptions, and the match between the two treatments. The number 1 for the CDQN treatment result for C1 means that CDQN recommended action number 1 (MET) for C1 patients with State 1. Similar to C2, for State 25, CDQN recommended action number 4 (MET + SUL). Lastly, CDQN recommended action number 7 (MET + SUL + AGI) for State 72. The experts also prescribed the same medication as the proposed CDQN for C1, C2, and C3 patients in State 1, State 25, and State 72, respectively. Therefore, we can conclude that the CDQN recommendation and experts' prescription matched.

Each clusters holds their respective state numbers according to the condition of patients in the group. State numbers 1 to 36 contain states with no diabetes complication and state numbers 37 to 72 contains states with diabetes complication. This is the reason that C3 has no recommendation result in State 1 to 36 because patients in C3 has more than one diabetes complication, whereas patients in C1 and C2 do not. Therefore, C1 and C2 relate to States 1 to 36 and C3 relates to States 37 to 72. The C1 take the minor T2DM state (16 states) and C2 take the severe T2DM state (20 states) among state 1 to 36 and the states of C1 and C2 does not overlap as shown in [Table 10](#). This is because we grouped the patients by their characteristics and attempted to recommend appropriate medication for each group.

When we observe the states belong to C2, it shows that there are more than 2 severe T2DM characteristics exist such as "Diabetes" state of FPG level, "8 to 11 years" of T2DM period,

Table 10
Sample state division of C1 and C2 among state 1 to 36.

State	Age	FPG	Complications	Period	Risk	Cluster
1	0	0	0	0	0	C1
2	0	0	0	1	0	C1
4	0	1	0	0	0	C1
25	1	2	0	0	0	C2
26	1	2	0	1	0	C2
28	1	0	0	0	1	C2

Table 11
Concordance rate between CDQN's recommendation and experts' prescription.

Clusters	No. of matched states	Concordance rate
C1	13/16	81.25%
C2	17/20	85.00%
C3	30/36	83.33%

more than age "55 years old", and "With" diabetes risk which are highlighted in red while C1 shows minor T2DM characteristics.

The concordance rate results for the three clusters are presented in [Table 11](#). The concordance rate results demonstrate that C1, C2, and C3 had concordance rates of 81.25%, 85.00%, and 83.33%, respectively. We can state that the proposed CDQN model is adequate for supporting T2DM treatment recommendations because all clusters primarily matched the real-life experts' prescriptions; certain medications could differ based on the state of the patient and clinician's priority.

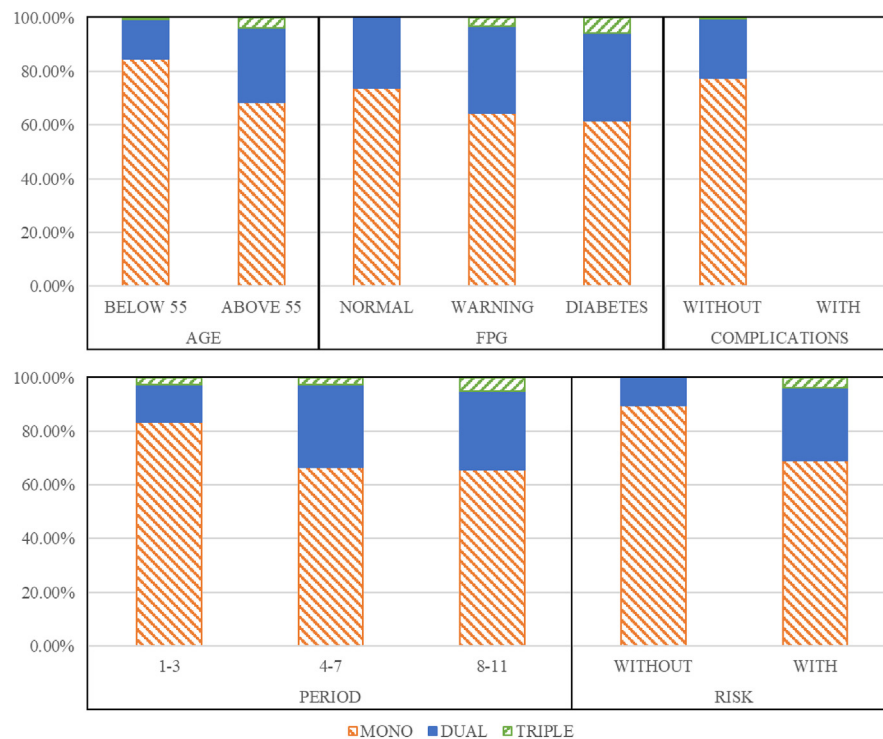


Fig. 3. T2DM treatment recommendation result of C1 by state components.

3.3. T2DM treatment recommendation results of each cluster by state

The results of the CDQN recommended T2DM treatment for each cluster are presented in this section. To observe trends in the use of utilizing multiple medications, we organized the recommendations based on the number of medications. Fig. 3 displays the medication recommended for each state component in C1 patients.

Fig. 3 displays the recommended treatment for C1 patients based on the number of medications. As C1 consists of minor symptoms in T2DM patients, we can observe that the proportion of monotherapy recommendations was greater than that of dual and triple therapy. Monotherapy decreased by 16% when patients reached 55 years of age; dual and triple therapy increased by 13% and 3%, respectively. Most importantly, as the severity of the FPG level worsened, the percentage of dual and triple therapy increased by 6% and 5%, respectively, when compared to the Normal to Diabetic state. Patients in C1 do not have diabetes-related complications; therefore, medication was recommended for only the state without complications, consisting of 77% monotherapy, 22% dual therapy, and 1% triple therapy. The recommendation trends in the period state indicated patterns similar to those of other states, which indicated that dual and triple therapies increased by 16% and 3%, respectively, as the period increased. Finally, patients with diabetes risk were recommended more dual and triple therapies by 17% and 4%, respectively, than patients without diabetes risk. Based on these results, we can verify that the recommended medications transit from monotherapy to dual or triple therapy as the health conditions of patients change. For concrete verification, we also analyzed the results of C2 and C3 in Figs. 4 and 5.

Fig. 4 displays the recommended treatment for C2 patients. Patients in C2 consisted of those with severe T2DM. Unlike the results in Fig. 3, we can observe that the proportion of dual and triple therapy recommendations was greater than that of

monotherapy. However, in C2, we can also observe the recommendation trend that moved from monotherapy to dual or triple therapy as the health condition of the patients worsened. In the age state, dual and triple therapy increased by 12% and 7%, respectively, when patients reach the age of 55 years. When the FPG level state moved from Normal to Diabetes, the dual and triple therapy recommendations increased by 16% and 8%, respectively. Patients in C2 also do not have diabetes-related complications; therefore, medication is recommended for only the state without complications, consisting of 51% monotherapy, 44% dual therapy, and 5% triple therapy. The recommendation in the period state also followed the trend of increase in dual and triple therapy as the period became longer. Lastly, patients with diabetes risk were recommended 11% and 10% more dual and triple therapy, respectively, than patients without diabetes risk.

Fig. 5 displays the recommended treatment for C3 patients. Patients in C3 consist of those with severe T2DM and diabetes complications. Similar to Fig. 4, we can observe that the proportion of dual and triple therapy recommendations was greater than that of monotherapy. In the age state, dual and triple therapy increased by 12% and 4%, respectively, when patients reach the age of 55 years. When the FPG level state moved from Normal to Diabetes, the dual and triple therapy recommendations increased by 6% and 9%, respectively. The important part in C3 is the complications state, compared to C1 and C2, which do not have complications. Recommendations in C3 indicate that more than half of the recommendations consisted of dual and triple therapy for patients with complications. The period state followed the trend of an increase in dual and triple therapy as the period increased. Lastly, patients with diabetes risk were recommended 13% and 10% more dual and triple therapy, respectively, than patients without diabetes risk. With the results from C1, C2, and C3, we can clearly conclude that the recommendation trend moves from monotherapy to dual or triple therapy as the health state of patients becomes more severe.

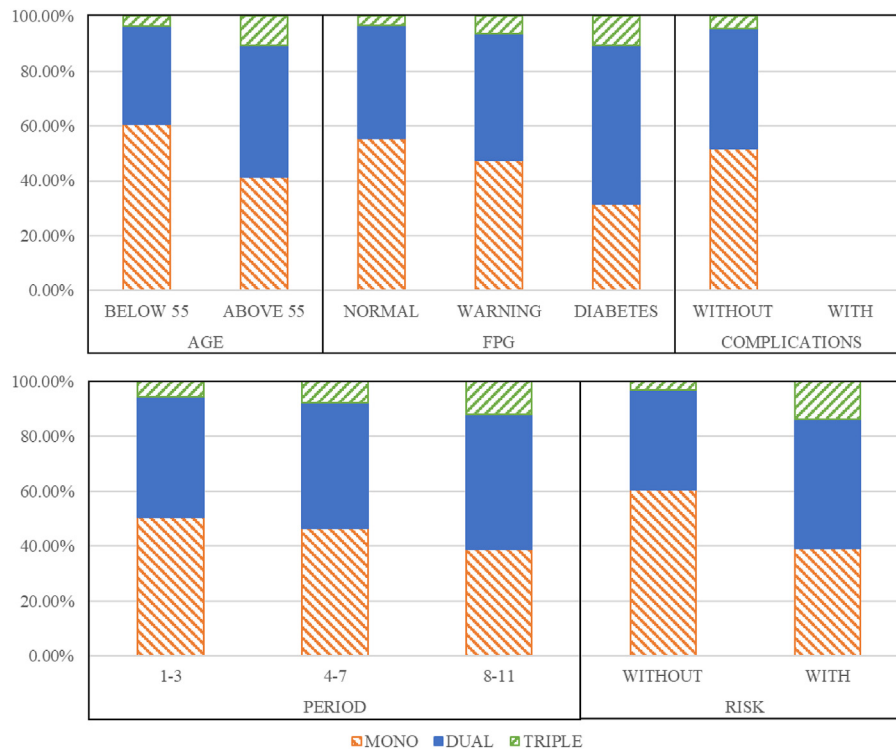


Fig. 4. T2DM treatment recommendation result of C2 by state components.

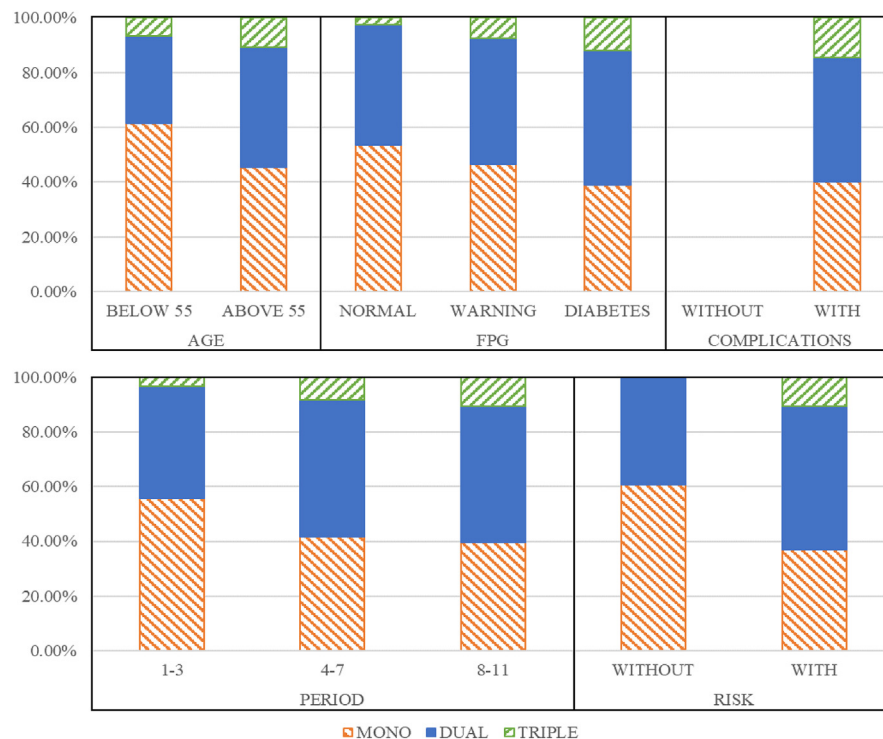


Fig. 5. T2DM treatment recommendation result of C3 by state components.

3.4. Medication possession ratio (MPR) validation

To validate adherence, the MPR of CDQN-prescribed medication was calculated. Medication adherence [54] refers to how well a patient takes the medications prescribed by his or her

doctor. Previously, the proportion of time when a pharmaceutical supply was available was used to define MPR-related adherence measures [55–57]. (7) can be used to compute the MPR [54].

$$\text{MPR} = \frac{\text{Days received medication}}{\text{Total days in period}} \times 100 \quad (7)$$

Table 12
Mean MPR results of clusters.

Clusters	Monotherapy	Dual therapy	Triple therapy
C1	85.20%	88.49%	81.09%
C2	85.75%	84.93%	83.01%
C3	87.39%	86.57%	83.56%

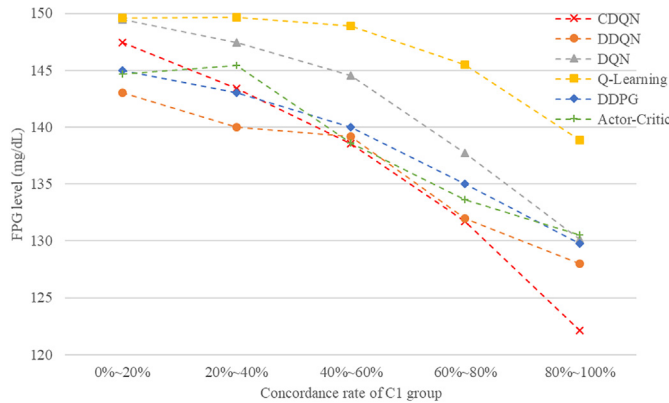


Fig. 6. Concordance rate vs. FPG level of C1 group.

We used the MPR to track treatment compliance with the proposed CDQN-recommended medications to confirm that following the recommendations improved patients' health. Table 12 presents the mean MPR results for each cluster.

The results indicate that the mean MPR for all medications in every cluster exceeded 81.09%. Based on these results, we can confirm that patients in all clusters complied with the CDQN recommended medication.

4. Performance comparison with existing RL models

In this section, we compare the CDQN recommendation with other existing RL methods such as Q-learning [22], DQN [58], DDQN [59], DDPG [25], and Actor-Critic [23] to verify the performance of the proposed method. The comparison metrics were the FPG level and diabetes-related complication occurrence rate based on the concordance rate of each RL model.

4.1. Concordance rate vs. FPG level

We used the FPG level variation to validate the performance of the four RL models. Figs. 6, 7, and 8 indicate the relationship between the patient concordance rate and FPG level for C1, C2, and C3, respectively. Based on the model-concordant rate, the patients were divided into 20% groups and the average FPG level in each group was computed.

In Fig. 6, the curves of all models indicate a decreasing trend. At the 0%–20% point of concordance rate, the FPG levels were in the highest position; however, as the patients began to adhere to the model, the FPG level decreased. This means that the models recommend appropriate medications to patients to maintain good health. The FPG levels of Q-learning, DQN, DDQN, DDPG, and actor-critic decreased by 7.20%, 10.29%, 11.16%, 10.48%, and 9.13%, respectively. The proposed CDQN model decreased the FPG level by 17.19%, which demonstrated the best performance among all RL methods.

The curves in Fig. 7 also indicate a decreasing trend. Because the patients in C2 had high FPG levels compared to those in C1, their average FPG level at the portion of concordance rate 0%–20% starts from approximately 165 mg/dL. This indicates FPG levels in the severe condition, displaying the highest level. Then, as the

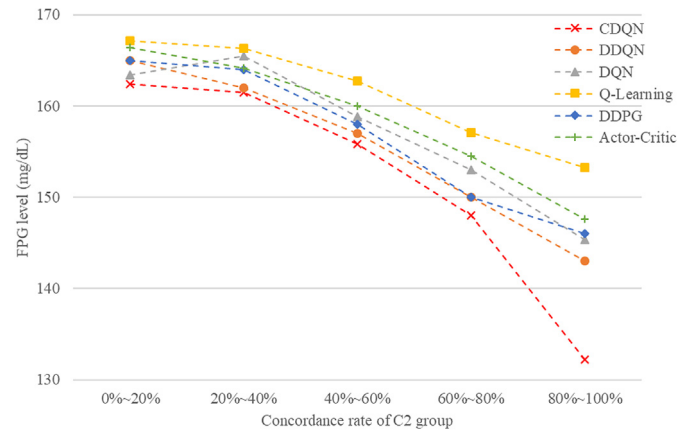


Fig. 7. Concordance rate vs. FPG level of C2 group.

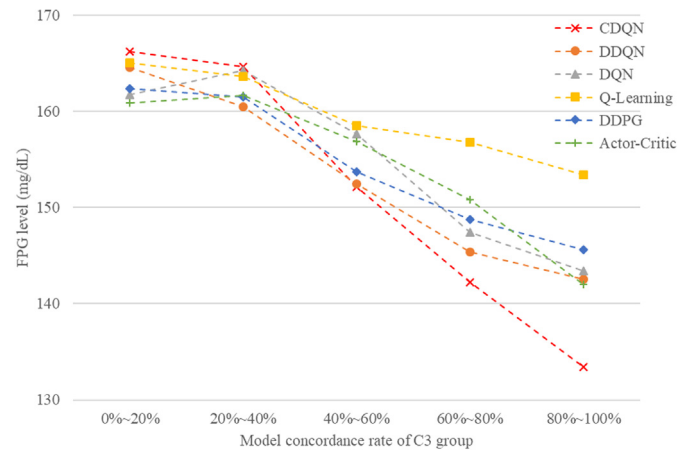


Fig. 8. Concordance rate vs. FPG level of C3 group.

patients begin to agree with the model, the FPG level decreased. Following the treatment recommendations of each model, the FPG levels of Q-learning, DQN, DDQN, DDPG, and actor-critic decreased by 8.44%, 11.06%, 13.24%, 11.51%, and 11.28%, respectively. The proposed CDQN model provided the best performance among all RL methods by decreasing the FPG level by 18.58%.

Finally, as indicated in Fig. 8, all RL models decreased the FPG level when patients were concordant with the models. The FPG levels of Q-learning, DQN, DDQN, DDPG, and actor-critic decreased by 7.05%, 11.32%, 13.36%, 10.35%, and 11.76%, respectively. The proposed CDQN model decreased the FPG level by 19.75%, which demonstrated the best performance among all RL methods. Based on these results, we can confirm that the CDQN model recommends appropriate medications, and by following the recommendation of the CDQN model, patients can maintain better health conditions. Therefore, the performance of the CDQN model was superior to that of other existing RL models.

4.2. Concordance rate vs. Occurrence rate of diabetes-related complication

We evaluated the performance of the proposed CDQN model and other existing RL models by observing the occurrence rate of diabetes-related complications. In this verification, only C3 was used because patients in C1 and C2 did not have diabetes complications. Based on the concordance rate, the patients were divided into 20% groups and the average diabetes complication occurrence rate in each group was computed. Fig. 9 displays the

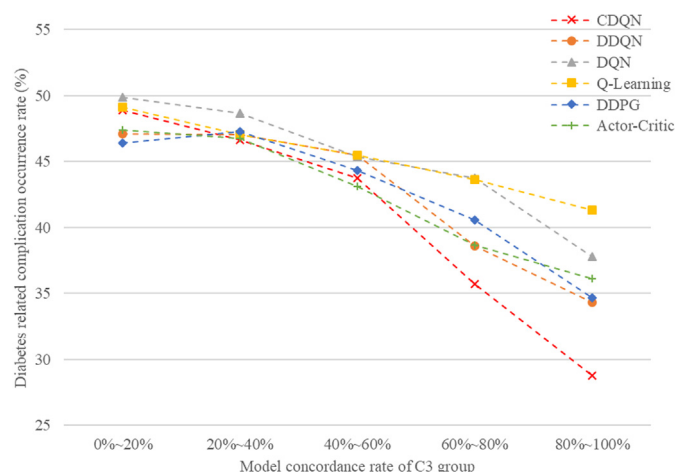


Fig. 9. Concordance rate vs. Diabetes-related complication occurrence rate of C3 group.

relationship between the concordance and occurrence rates of diabetes-related complications.

In Fig. 9, the curves of each model decrease as the concordance rate increases. Similar to the FPG level, when a patient's concordance rate is low, the diabetes-related complication occurrence rate is at a high risk; however, as the patient becomes adherent to the model, the complication occurrence rate decreases. The diabetes-related complication occurrence rates of Q-learning, DQN, DDQN, DDPG, and actor-critic decreased by 7.77%, 12.10%, 12.77%, 11.76%, and 11.22%, respectively. The proposed CDQN model decreased the complication occurrence rate by 20.11%, which was the best performance among the RL models. Once again, we can confirm that the CDQN model recommends proper medications, and by following the recommendation of the CDQN model, the patients can reduce the risk of complications.

5. Conclusions

In this study, we proposed a cluster-applied deep RL-based T2DM treatment recommendation model. To design the model, we used EHRs from the South Korean database. Our purpose was to apply a k-means clustering algorithm to group patients who are in similar states to boost the performance of the deep RL and build a more realistic treatment recommendation model that could support clinical experts in improving research in the field of healthcare. We selected the state component using age, FPG, complications, T2DM period, and diabetes risk to include detailed information on patients with T2DM. Moreover, we considered eight possible medications for diabetes treatment.

Our main contribution is that we designed an enhanced T2DM treatment recommendation model by applying the clustering algorithm to deep RL and considering essential factors for controlling diabetes, such as demographic information, body measurements, and laboratory test values for state components. Based on our defined patient state, the CDQN model could recommend T2DM treatments as mono, dual, or triple therapy. We validated the proposed model's recommendations by comparing them with real-life expert prescriptions. Moreover, with the results of the treatment recommendations for the three clusters, we found that as patients' conditions worsened, the proposed model tended to recommend a greater number of medications. This was verified by the results wherein the minor T2DM group (C1) had more recommendations for monotherapy than the severe T2DM group (C2) and T2DM with complications group (C3), which had a greater proportion of dual and triple therapy recommendations.

Secondly, we performed MPR validation to verify the patients' adherence and compliance with the proposed CDQN recommended medication because the effective control of T2DM treatment is also associated with the adherence of patients to medication. As the average MPR for all therapies of every cluster was above 81%, we can assume that the patients in each cluster complied well with our recommended medications. After confirming that the recommendations of the proposed model were accepted and followed by patients, we examined T2DM management performance by observing the correlation between the patient's model-concordance rate and variation in FPG level and occurrence rate of diabetes-related complications.

Thirdly, we demonstrated the performance of the proposed CDQN model by comparing it with other existing RL models such as Q-learning, DQN, DDQN, DDPG, and actor-critic. The comparison metrics were the correlation between the patient's model-concordance rate and variation in FPG level and the occurrence rate of diabetes-related complications. The outcome revealed that the model-concordance rate and FPG level, and the occurrence rate of diabetes complications have a negative relationship, which means that the higher the patient's model-concordance rate, the lower the FPG level and the lower the occurrence rate of diabetes complications. Therefore, we confirmed that the proposed CDQN model is effective. Moreover, the proposed CDQN outperformed all other existing RL models.

In conclusion, with respect to all the results and validations, we proved that the proposed cluster-applied deep RL-based T2DM treatment recommendation model could recommend appropriate medications and help in effective T2DM management. Moreover, we verified that applying a clustering algorithm to a deep RL model can increase the learning performance by demonstrating the proposed CDQN model's outstanding performance compared to other existing RL models.

6. Limitations and future works

Our study has a limitation in that it only analyzed one type of EHR offered by the Korean institute. However, this could be an advantage of our research because only a limited number of studies in the field of healthcare and deep learning have used EHRs from South Koreans. Nevertheless, it would be better to have more medical databases to compare the results among races and ethnicities. Moreover, we did not use the HbA1c level, which is a crucial factor for the diagnosis of diabetes, because of its unavailability in the database. Instead, we used FPG values to indicate the patient's diabetes state, as they can also be used to check the diabetes level.

In future studies, we intend to obtain additional EHRs by cooperating with other Korean healthcare institutions or networks from different nations to verify and compare the outcomes of the proposed model. Furthermore, integrating different databases could allow us to broaden the disease area. In this study, we only focused on patients with T2DM; however, we aim to expand this to additional diseases in the future.

Finally, in clinical research, it is vital to manage prescription uncertainties. To become closer to a realistic and precise model that leads to successful recommendations, the proposed model analyzes and learns from numerous iterations in the database. Consequently, we believe that clinicians could employ data-driven deep-learning results to enhance clinical practice in their research.

CRedit authorship contribution statement

Sang Ho Oh: Conceptualization, Data curation, Methodology, Validation, Formal analysis, Writing – original draft, Visualization. **Su Jin Lee:** Conceptualization, Validation. **Jongyoul Park:** Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This study was supported by the Research Program funded by the SeoulTech (Seoul National University of Science and Technology). The funders had no role in study design, data collection, and analysis, decision to publish, or preparation of the manuscript.

Informed consent statement

Patient consent was waived due to the research type that deals with only gathered data from National Health Insurance Sharing Service. The authors do not collect any other additional patient information.

References

- [1] T.H. Su, C.H. Wu, J.H. Kao, Artificial intelligence in precision medicine in hepatology, *J. Gastroenterol. Hepatol.* 36 (2021) 569–580, <http://dx.doi.org/10.1111/jgh.15415>.
- [2] S.J. MacEachern, N.D. Forkert, Machine learning for precision medicine, *Genome* 64 (2021) 416–425, <http://dx.doi.org/10.1139/gen-2020-0131>.
- [3] M. van der Schaar, A.M. Alaa, A. Floto, A. Gimson, S. Scholtes, A. Wood, E. McKinney, D. Jarrett, P. Lio, A. Ercole, How artificial intelligence and machine learning can help healthcare systems respond to COVID-19, *Mach. Learn.* 110 (2020) 1–14, <http://dx.doi.org/10.1007/s10994-020-05928-x>.
- [4] F. Jiang, Y. Jiang, H. Zhi, Y. Dong, H. Li, S. Ma, Y. Wang, Q. Dong, H. Shen, Y. Wang, Artificial intelligence in healthcare: past, present and future, *Stroke Vasc. Neurol.* 2 (2017) 230–243, <http://dx.doi.org/10.1136/svn-2017-000101>.
- [5] G. Rong, A. Mendez, E.B. Bou Assi, B. Zhao, M. Sawan, Artificial intelligence in healthcare: review and prediction case studies, *Engineering* 6 (2020) 291–301, <http://dx.doi.org/10.1016/j.eng.2019.08.015>.
- [6] D. Lee, S.N. Yoon, Application of artificial intelligence-based technologies in the healthcare industry: opportunities and challenges, *Int. J. Environ. Res. Public Health* 18 (2021) 271, <http://dx.doi.org/10.3390/ijerph18010271>.
- [7] S.H. Oh, S.J. Lee, J. Park, Precision medicine for hypertension patients with type 2 diabetes via reinforcement learning, *J. Pers. Med.* 12 (2022) 87, <http://dx.doi.org/10.3390/jpm12010087>.
- [8] A. Nayarisseri, R. Khandelwal, P. Tanwar, M. Madhavi, D. Sharma, G. Thakur, A. Speck-Planche, S.K. Singh, Artificial intelligence, big data and machine learning approaches in precision medicine & drug discovery, *Curr. Drug Targets* 22 (2021) 631–655, <http://dx.doi.org/10.2174/1389450122999210104205732>.
- [9] F. Piccialli, F. Calabrò, D. Crisci, S. Cuomo, E. Prezioso, R. Mandile, R. Troncone, L. Greco, R. Auricchio, Precision medicine and machine learning towards the prediction of the outcome of potential celiac disease, *Sci. Rep.* 11 (2021) 5683, <http://dx.doi.org/10.1038/s41598-021-84951-x>.
- [10] C. Lam, A. Siefkas, N.S. Zelin, G. Barnes, R.P. Dellinger, J.L. Vincent, G. Braden, H. Burdick, J. Hoffman, J. Calvert, Q. Mao, R. Das, Machine learning as a precision-medicine approach to prescribing covid-19 pharmacotherapy with remdesivir or corticosteroids, *Clin. Ther.* 43 (2021) 871–885, <http://dx.doi.org/10.1016/j.clinthera.2021.03.016>.
- [11] T. Infante, M. Franccone, M.L. De Rimini, C. Cavaliere, R. Canonico, C. Catalano, C. Napoli, Machine learning and network medicine: a novel approach for precision medicine and personalized therapy in cardiomyopathies, *J. Cardiovasc. Med. (Hagerstown)* 22 (2021) 429–440, <http://dx.doi.org/10.2459/JCM.0000000000001103>.
- [12] A. Coronato, M. Naeem, G. De Pietro, G. Paragliola, Reinforcement learning for intelligent healthcare applications: A survey, *Artif. Intell. Med.* 109 (2020) 101964, <http://dx.doi.org/10.1016/j.artmed.2020.101964>.
- [13] C. Yu, J. Liu, S. Nemati, G. Yin, Reinforcement learning in healthcare: A survey, *ACM Comput. Surv. (CSUR)* 55 (2021) 1–36, <http://dx.doi.org/10.1145/3477600>.
- [14] C. Langenberg, L.A. Lotta, Genomic insights into the causes of type 2 diabetes, *Lancet* 391 (2018) 2463–2474, [http://dx.doi.org/10.1016/S0140-6736\(18\)31132-2](http://dx.doi.org/10.1016/S0140-6736(18)31132-2).
- [15] N. Kaul, S. Ali, Genes, genetics, and environment in type 2 diabetes: implication in personalized medicine, *DNA Cell Biol.* 35 (2016) 1–12, <http://dx.doi.org/10.1089/dna.2015.2883>.
- [16] N.H. Cho, J.E. Shaw, S. Karuranga, Y. Huang, J.D. da Rocha Fernandes, A.W. Ohlrogge, B. Malanda, IDF diabetes atlas: global estimates of diabetes prevalence for 2017 and projections for 2045, *Diabetes Res. Clin. Pract.* 138 (2018) 271–281, <http://dx.doi.org/10.1016/j.diabres.2018.02.023>.
- [17] A. Artasensi, A. Pedretti, G. Fumagalli, Type 2 diabetes mellitus: a review of multi-target drugs, *Molecules* 25 (2020) <http://dx.doi.org/10.3390/molecules25081987>.
- [18] M. Haghighatpanah, A.S.M. Nejad, M. Haghighatpanah, G. Thunga, S. Malayasamy, Factors that correlate with poor glycemic control in type 2 diabetes mellitus patients with complications, *Osong Public Health Res. Perspect.* 9 (2018) 167–174, <http://dx.doi.org/10.24171/j.phrp.2018.9.4.05>.
- [19] A.A. Kakade, I.R. Mohanty, S. Rai, Assessment of factors associated with poor glycemic control among patients with Type II Diabetes mellitus, *Integr. Obes. Diabetes* 4 (2018) <http://dx.doi.org/10.15761/iod.1000209>.
- [20] M. Khattab, Y.S. Khader, A. Al-Khawaldeh, K. Ajlouni, Factors associated with poor glycemic control among patients with type 2 diabetes, *J. Diabetes Complicat.* 24 (2010) 84–89, <http://dx.doi.org/10.1016/j.jdiacomp.2008.12.008>.
- [21] O. Gottesman, F. Johansson, M. Komorowski, A. Faisal, D. Sontag, F. Doshi-Velez, L.A. Celi, Guidelines for reinforcement learning in healthcare, *Nat. Med.* 25 (2019) 16–18, <http://dx.doi.org/10.1038/s41591-018-0310-5>.
- [22] M.O.M. Oroojeni Mohammad Javad, S.O. Agboola, K. Jethwani, A. Zeid, S. Kamathi, A reinforcement learning-based method for management of type 1 diabetes: exploratory study, *JMIR Diabetes* 4 (2019) e12905, <http://dx.doi.org/10.2196/12905>.
- [23] G. Schamberg, M. Badgeley, B. Meschede-Krasa, O. Kwon, E.N. Brown, Continuous action deep reinforcement learning for propofol dosing during general anesthesia, *Artif. Intell. Med.* 123 (2022) 102227, <http://dx.doi.org/10.1016/j.artmed.2021.102227>.
- [24] H. Zheng, I.O. Ryzhov, W. Xie, J. Zhong, Personalized multimorbidity management for patients with type 2 diabetes using reinforcement learning of electronic health records, *Drugs* 81 (2021) 471–482, <http://dx.doi.org/10.1007/s40265-020-01435-4>.
- [25] T. Lin, X. Zhang, J. Gong, R. Tan, X. Xu, L. Wang, et al., A dosing strategy model of deep deterministic policy gradient algorithm for sepsis, 2022, <http://dx.doi.org/10.21203/rs.3.rs-1391838/v1>.
- [26] X. Sun, Y.M. Bee, S.W. Lam, Z. Liu, W. Zhao, S.Y. Chia, H. Abdul Kadir, J.T. Wu, B.Y. Ang, N. Liu, Z. Lei, Z. Xu, T. Zhao, G. Hu, G. Xie, Effective treatment recommendations for Type 2 diabetes management using reinforcement learning: treatment recommendation model development and validation, *J. Med. Internet Res.* 23 (2021) e27858, <http://dx.doi.org/10.2196/27858>.
- [27] N.M. Maruthur, E. Tseng, S. Hutfless, L.M. Wilson, C. Suarez-Cuervo, Z. Berger, Y. Chu, E. Iyoha, J.B. Segal, S. Bolen, Diabetes medications as monotherapy or metformin-based combination therapy for type 2 diabetes: a systematic review and meta-analysis, *Ann. Intern. Med.* 164 (2016) 740–751, <http://dx.doi.org/10.7326/M15-2650>.
- [28] UK Prospective Diabetes Study (UKPDS) Group, Effect of intensive blood-glucose control with metformin on complications in overweight patients with type 2 diabetes (UKPDS 34), *Lancet* 352 (1998) 854–865, [http://dx.doi.org/10.1016/S0140-6736\(98\)07037-8](http://dx.doi.org/10.1016/S0140-6736(98)07037-8).
- [29] C.M.Y. Lee, M. Woodward, S. Colagiuri, Triple therapy combinations for the treatment of type 2 diabetes—a network meta-analysis, *Diabetes Res. Clin. Pract.* 116 (2016) 149–158, <http://dx.doi.org/10.1016/j.diabres.2016.04.037>.
- [30] D. Qian, T. Zhang, P. Zheng, Z. Liang, S. Wang, J. Xie, L. Zhao, Y. Zhang, B. Situ, Comparison of oral antidiabetic drugs as add-on treatments in patients with type 2 diabetes uncontrolled on metformin: a network meta-analysis, *Diabetes Ther.* 9 (2018) 1945–1958, <http://dx.doi.org/10.1007/s13300-018-0482-5>.
- [31] F. Zaccardi, N.N. Dhalwani, J. Dales, H. Mani, K. Khunti, M.J. Davies, D.R. Webb, Comparison of glucose-lowering agents after dual therapy failure in type 2 diabetes: a systematic review and network meta-analysis of randomized controlled trials, *Diabetes Obes. Metab.* 20 (2018) 985–997, <http://dx.doi.org/10.1111/dom.13185>.
- [32] L. Olansky, C. Reasner, T.L. Seck, D.E. Williams-Herman, M. Chen, L. Terranella, A. Mehta, K.D. Kaufman, B.J. Goldstein, A treatment strategy implementing combination therapy with sitagliptin and metformin results in superior glycaemic control versus metformin monotherapy due to a low rate of addition of antihyperglycaemic agents, *Diabetes Obes. Metab.* 13 (2011) 841–849, <http://dx.doi.org/10.1111/j.1463-1326.2011.01416.x>.
- [33] S. Hadjadj, J. Rosenstock, T. Meinicke, H.J. Woerle, U.C. Broedl, Initial combination of empagliflozin and metformin in patients with type 2 diabetes, *Diabetes Care* 39 (2016) 1718–1728, <http://dx.doi.org/10.2337/dc16-0522>.
- [34] G. Lozano-Ortega, S. Goring, H.A. Bennett, K. Bergenheim, C. Sternhufvud, J. Mukherjee, Network meta-analysis of treatments for type 2 diabetes mellitus following failure with metformin plus sulfonylurea, *Curr. Med. Res. Opin.* 32 (2016) 807–816, <http://dx.doi.org/10.1185/03007995.2015.1135110>.

- [35] X. Cai, X. Gao, W. Yang, X. Han, L. Ji, Efficacy and safety of initial combination therapy in treatment-naïve type 2 diabetes patients: a systematic review and meta-analysis, *Diabetes Ther.* 9 (2018) 1995–2014, <http://dx.doi.org/10.1007/s13300-018-0493-2>.
- [36] C. Yuan, H. Yang, Research on K-value selection method of K-means clustering algorithm, *Joural* 2 (2019) 226–235, <http://dx.doi.org/10.3390/j2020016>.
- [37] J. Wu, Cluster analysis and K-means clustering: An introduction, in: *Advances in K-Means Clustering*, 2012, pp. 1–16, http://dx.doi.org/10.1007/978-3-642-29807-3_1.
- [38] R.R. Rathod, R.D. Garg, Design of electricity tariff plans using gap statistic for K-means clustering based on consumers monthly electricity consumption data, *Int. J. Energy Sect. Manage.* 11 (2017) 295–310, <http://dx.doi.org/10.1108/IJESM-02-2016-0005>.
- [39] D. Arthur, S. Vassilvitskii, K-means++: The advantages of careful seeding, in: *Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms*, Society for Industrial and Applied Mathematics, 2007, pp. 1027–1035.
- [40] S. Hjerde, Evaluating Deep Q-Learning Techniques for Controlling Type 1 Diabetes Masters, *UiT Norges arktiske universitet*, 2020.
- [41] A. Nair, P. Srinivasan, S. Blackwell, C. Alciçek, R. Fearon, A. De Maria, et al., Massively parallel methods for deep reinforcement learning, 2020, <http://dx.doi.org/10.48550/arXiv.1507.04296>, arXiv preprint [arXiv:1507.04296](https://arxiv.org/abs/1507.04296).
- [42] R. Kumar, R.K. Aggarwal, J.D. Sharma, Energy analysis of a building using artificial neural network: a review, *Energy Build.* 65 (2013) 352–358, <http://dx.doi.org/10.1016/j.enbuild.2013.06.007>.
- [43] D. Kingma, J. Ba, Adam: A method for stochastic optimization, 2014, <http://dx.doi.org/10.48550/arXiv.1412.6980>, arXiv preprint [arxiv:1412.6980](https://arxiv.org/abs/1412.6980).
- [44] V. Mnih, K. Kavukcuoglu, D. Silver, A.A. Rusu, J. Veness, M.G. Bellemare, A. Graves, M. Riedmiller, A.K. Fidjeland, G. Ostrovski, S. Petersen, C. Beattie, A. Sadik, I. Antonoglou, H. King, D. Kumaran, D. Wierstra, S. Legg, D. Hassabis, Human-level control through deep reinforcement learning, *Nature* 518 (2015) 529–533, <http://dx.doi.org/10.1038/nature14236>.
- [45] K.U. Ahn, C.S. Park, Application of deep Q-networks for model-free optimal control balancing between different HVAC systems, *Sci. Technol. Built Environ.* 26 (2020) 61–74, <http://dx.doi.org/10.1080/23744731.2019.1680234>.
- [46] E. Wong, K. Backholer, E. Gearon, J. Harding, R. Freak-Poli, C. Stevenson, A. Peeters, Diabetes and risk of physical disability in adults: a systematic review and meta-analysis, *Lancet Diabetes Endocrinol.* 1 (2013) 106–114, [http://dx.doi.org/10.1016/S2213-8587\(13\)70046-9](http://dx.doi.org/10.1016/S2213-8587(13)70046-9).
- [47] S.J. Griffin, P.S. Little, C.N. Hales, A.L. Kinmonth, N.J. Wareham, Diabetes risk score: towards earlier detection of type 2 diabetes in general practice, *Diabetes Metab. Res. Rev.* 16 (2000) 164–171, [http://dx.doi.org/10.1002/1520-7560\(200005/06\)16:3<164::aid-dmrr103>3.0.co;2-r](http://dx.doi.org/10.1002/1520-7560(200005/06)16:3<164::aid-dmrr103>3.0.co;2-r).
- [48] J. Lindström, J. Tuomilehto, The diabetes risk score: a practical tool to predict type 2 diabetes risk, *Diabetes Care* 26 (2003) 725–731, <http://dx.doi.org/10.2337/diacare.26.3.725>.
- [49] M.C. Weinstein, G. Torrance, A. McGuire, QALYs: the basics, *Value Health* 12 (Supplement 1) (2009) S5–S9, <http://dx.doi.org/10.1111/j.1524-4733.2009.00515.x>.
- [50] P.M. Clarke, A.M. Gray, A. Briggs, R.J. Stevens, D.R. Matthews, R.R. Holman, UKPDS 72 United Kingdom prospective diabetes study, cost-utility analyses of intensive blood glucose and tight blood pressure control in type 2 diabetes (UKpds 72), *Diabetologia* 48 (2005) 868–877, <http://dx.doi.org/10.1007/s00125-005-1717-3>.
- [51] T.O. Tengs, A. Wallace, One thousand health-related quality-of-life estimates, *Med. Care* 38 (2000) 583–637, <http://dx.doi.org/10.1097/00005650-200006000-00004>.
- [52] A.F. Cardoso, R. Cruz, P. Queirós, L. Santiago, C.F. Ribeiro, P. Ferreira, Assessment of health-related quality of life using the EQ-5D-3L in individuals with type 2 diabetes mellitus, *J. Diabetes Metab. Disord. Control* 3 (2016) 00064, <http://dx.doi.org/10.15406/jdmdc.2016.03.00064>.
- [53] X. Zhou, K.R. Siegel, B.P. Ng, S. Jawanda, K.K. Proia, X. Zhang, A.L. Albright, P. Zhang, Cost-effectiveness of diabetes prevention interventions targeting high-risk individuals and whole populations: a systematic review, *Diabetes Care* 43 (2020) 1593–1616, <http://dx.doi.org/10.2337/dci20-0018>.
- [54] C.M. Kozma, M. Dickson, A.L. Phillips, D.M. Meletiche, Medication possession ratio: implications of using fixed and variable observation periods in assessing adherence with disease-modifying drugs in patients with multiple sclerosis, *Patient Prefer. Adherence* 7 (2013) 509–516, <http://dx.doi.org/10.2147/PPA.S40736>.
- [55] S.E. Andrade, K.H. Kahler, F. Frech, K.A. Chan, Methods for evaluation of medication adherence and persistence using automated databases, *Pharmacoepidemiol. Drug Saf.* 15 (2006) 565–574, <http://dx.doi.org/10.1002/pds.1230>, discussion 575.
- [56] J.A. Cramer, A. Roy, A. Burrell, C.J. Fairchild, M.J. Fuldeore, D.A. Ollendorf, P.K. Wong, Medication compliance and persistence: terminology and definitions, *Value Health* 11 (2008) 44–47, <http://dx.doi.org/10.1111/j.1524-4733.2007.00213.x>.
- [57] E.L.P. Sattler, J.S. Lee, M. Perri, Medication (re) fill adherence measures derived from pharmacy claims data in older Americans: a review of the literature, *Drugs Aging* 30 (2013) 383–399, <http://dx.doi.org/10.1007/s40266-013-0074-z>.
- [58] V. Mnih, K. Kavukcuoglu, D. Silver, A. Graves, I. Antonoglou, D. Wierstra, et al., Playing atari with deep reinforcement learning, 2013, <http://dx.doi.org/10.48550/arXiv.1312.5602>, arxiv preprint [arxiv:1312.5602](https://arxiv.org/abs/1312.5602).
- [59] H. Van Hasselt, A. Guez, D. Silver, Deep reinforcement learning with double Q-learning, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 30, 2016, <http://dx.doi.org/10.1609/aaai.v30i1.10295>.